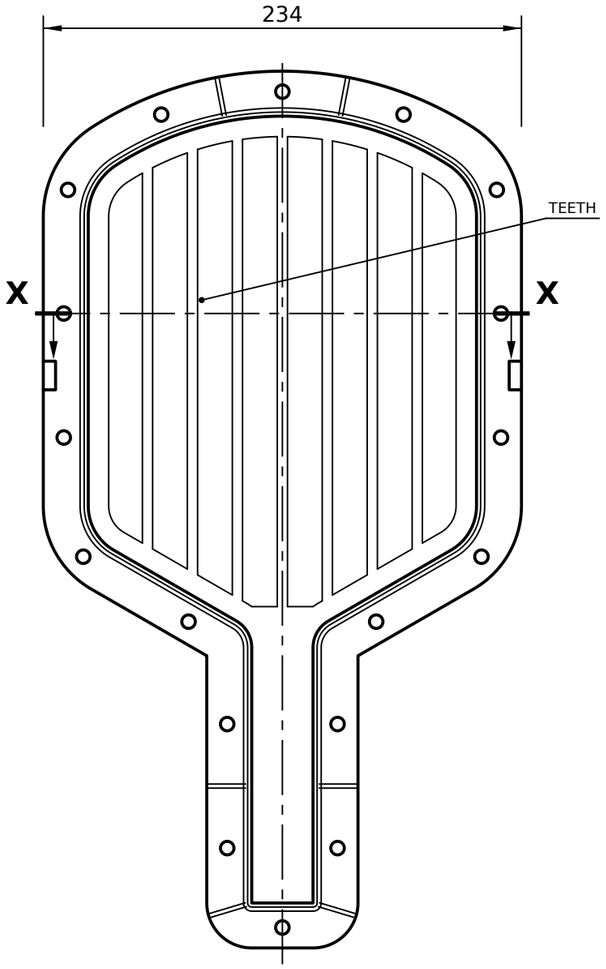
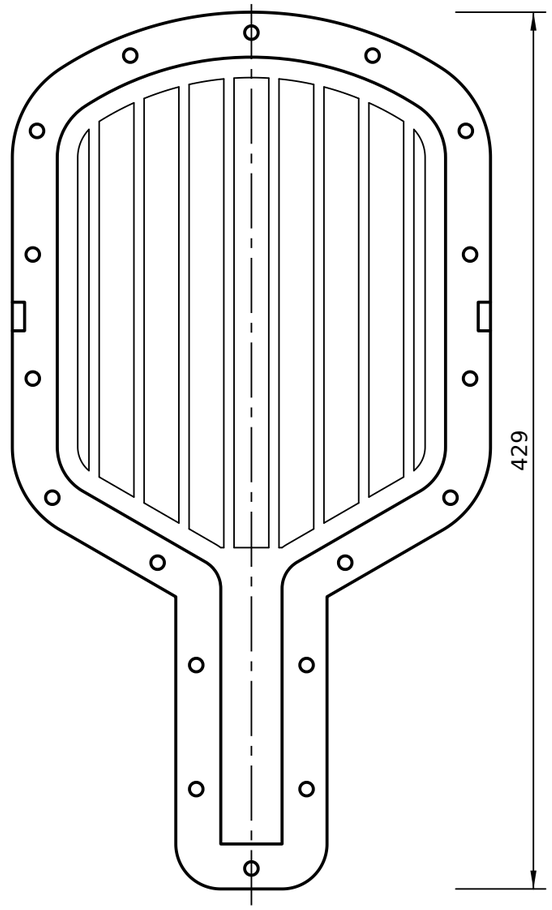


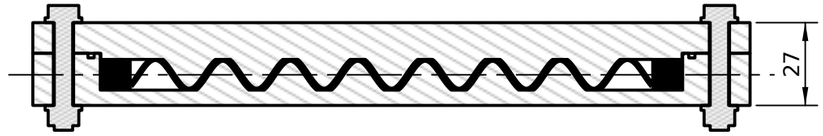
TRAY (BOTTOM HALF)

SCALE 1:3 · TEETH = BOTTOM FLUTES



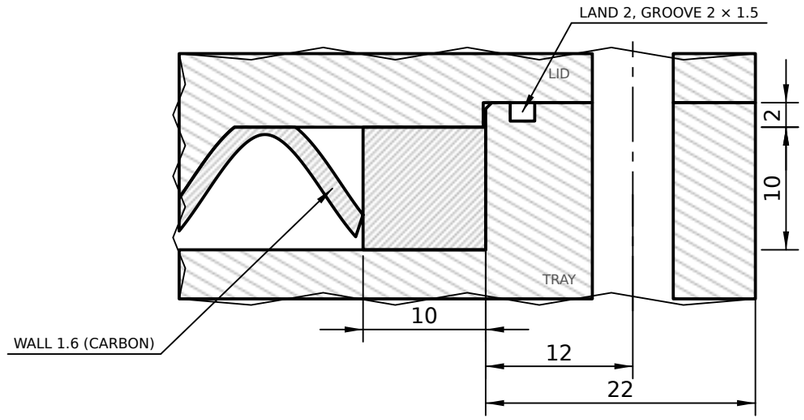
LID, UNDERSIDE

SCALE 1:3 · PLUG 2 HIGH · TEETH = TOP FLUTES



SECTION X–X

SCALE 1:2 · CLOSED MOULD, MOULDED RIB SUPPORT BLACK, M6 BOLT



DETAIL Y

SCALE 2:1 · PLUG IN THE POCKET, 0.2 GAP PER SIDE · THE GAPS ARE THE WALLS AND THE FRAME

NOTES

1. TRAY (1) + LID (2): output/mould/hybrid_mould_bottom / _top (.stl / .step). THE CAVITY BETWEEN THEM IS THE RIB SUPPORT (SHEET 3); THE TEETH ARE ITS FLUTES. CORE HEIGHT 10, WALLS 1.6.
2. THE LID'S 2 PLUG SLIDES INTO THE POCKET (0.2 CLEARANCE PER SIDE) AND ALIGNS THE HALVES; THE LID FLANGE SEATING ON THE TRAY RIM LEAVES THE PLUG FACE 10 ABOVE THE POCKET FLOOR.
3. PRINT BOTH BACK DOWN, TEETH UP: ≥ 5 WALLS, ≥ 40 % INFILL, 0.12–0.16 LAYERS FOR THE TEETH. NEEDS A ≥ 440 mm BED, OR MACHINE THE STEP FROM ALUMINIUM. ABOUT 0.9 + 1.1 kg OF PLA AT 100 %.
4. PLA SOFTENS NEAR 55 °C: ROOM-TEMPERATURE-CURE EPOXY ONLY. FOR A WARM CURE PRINT IN PETG, ASA OR PA-CF.
5. DRY FIT FIRST: THE PLUG MUST SLIDE IN BY HAND (SAND IT IF IT BINDS), AND ON THE TRIAL RIB SUPPORT STL PRINTED IN PLA THE FLANGE MUST SEAT FULLY. THEN SAND THE CAVITY (P400 → P800), SEAL, WAX ×3 + PVA.
6. FILL (FORGED CARBON) ≈ 301 g: HALF CHOPPED CARBON TOW (10–25 mm), HALF EPOXY BY MASS. PACK THE WALL GAPS AND THE FRAME FIRST, THEN THE SOLID HANDLE ZONE, AND SPREAD THE REST EVENLY. THE OPEN TRAY HOLDS 2.2× THE PART.
7. CLOSE: LID ON, BOLTS (3) FINGER-TIGHT, PRESS BETWEEN 2 FLAT, STIFF PLATENS OR IN A PRESS UNTIL THE FLANGE SEATS; TIGHTEN THE BOLTS. EXCESS RESIN RISES PAST THE PLUG INTO THE OVERFLOW GROOVE AND OUT OF THE VENTS.
8. CURE (24 h AT ROOM TEMPERATURE), SPLIT AT THE PRY NOTCHES, LIFT THE LID STRAIGHT UP, EASE THE PART OUT FROM THE BUTT END WITH A PLASTIC WEDGE (NO EJECTOR HOLES). TRIM THE FLASH, DRILL AND FIT THE M3 INSERT (SHEET 3). THE PART WEIGHS ≈ 274 g.

MOULD HARDWARE

ITEM	PART	QTY
1	Tray, hybrid_mould_bottom.stl	1
2	Lid, hybrid_mould_top.stl	1
3	Hex bolt M6 × 40, 2 washers, nut	18
4	Release wax + PVA; flat platens + clamps or a press	a/r

RIB SUPPORT MOULD – HYBRID

RIB SUPPORT MOULD · TRAY + LID WITH PLUG, TEETH INTERLOCK · 234 × 429 mm

DRAWN	DATE	CHECKED	APPROVED
Claude Code	2026-11-09		

DRAWING No.	REV	SHEET	SIZE	SCALE	UNITS
PB-HY-001	G	4 / 4	A3	1:3, 1:2, 2:1	mm

PROJECTION	MATERIAL	STATUS
	PLA / PETG (NOTE 2)	ISSUED FOR REVIEW